

KL Divergence: Comparing Two Probability Distributions

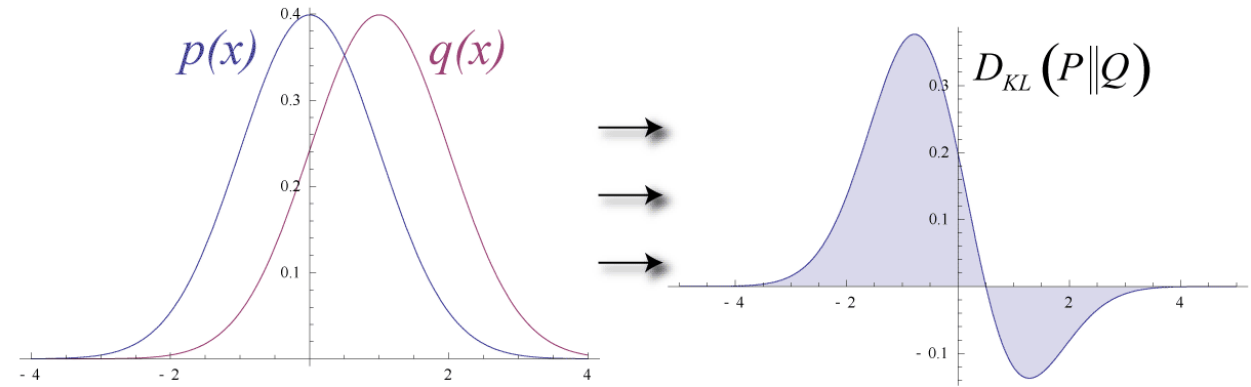
- The Kullback-Leibler (KL) divergence:
Measure the difference between two probability distributions over the same variable x
- From information theory, closely related to *relative entropy*, *information divergence*, and *information for discrimination*
- $D_{KL}(p(x) || q(x))$: divergence of $q(x)$ from $p(x)$, measuring the information lost when $q(x)$ is used to approximate $p(x)$

$$D_{KL}(p(x)||q(x)) = \sum_{x \in X} p(x) \ln \frac{p(x)}{q(x)}$$

Discrete form

$$D_{KL}(p(x)||q(x)) = \int_{-\infty}^{\infty} p(x) \ln \frac{p(x)}{q(x)} dx$$

Continuous form



Original Gaussian PDF's

KL Area to be Integrated

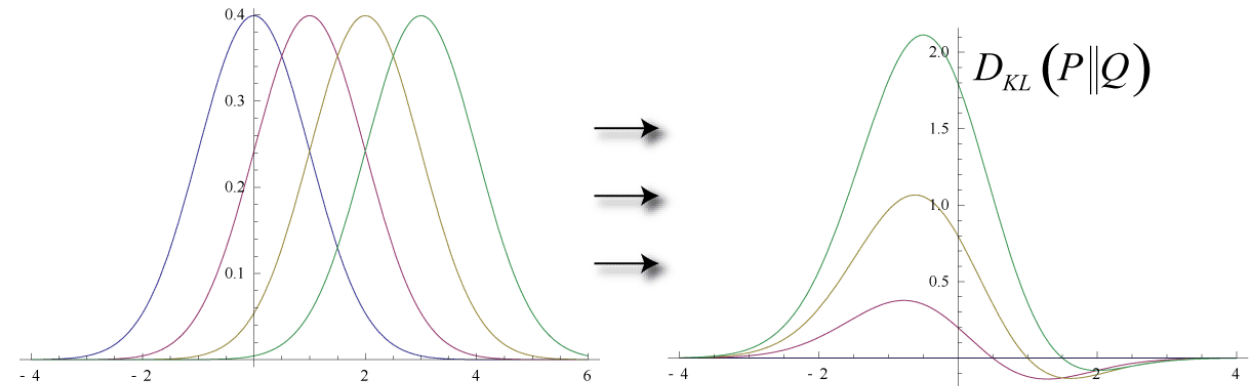


Image Source: The Kullback-Leibler (KL) divergence, Wikipedia